

Experiments with microwaves

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In this experiment, several fundamental properties of microwave radiation were investigated using a Gunn diode as a microwave source. The wavelength of the microwaves was determined using standing waves and a Michelson interferometer, yielding $\lambda = (3.256 \pm 0.052)cm$ and $\lambda = (3.300 \pm 0.100)cm$, respectively. Furthermore, the angular distribution and polarization properties of the radiation were examined. The polarization measurements confirmed that the emitted microwave radiation is linearly polarized. Frustrated total reflection was investigated using two paraffin prisms separated by an air gap. The transmitted intensity showed an exponential decrease with increasing prism separation, resulting in a penetration depth of $\delta = (2.31 \pm 0.14)cm$. In addition, Bragg diffraction on a macroscopic cubic crystal was used to determine the lattice constant, yielding values of

$d_0 = (3.74 \pm 0.14)cm$ and $d_0 = (4.03 \pm 0.10)cm$ for different lattice planes. Overall, the experiment demonstrated several characteristic wave phenomena of microwave radiation and illustrated their relevance for optics, solid state physics, and quantum mechanics.

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1 Introduction

The spectrum of electromagnetic radiation covers a wide range of wavelengths, ranging from high-energy γ -rays to low-energy radio waves, all of which play an important role in today's world. Examples include X-rays used in medical diagnostics, the radio waves employed by mobile phones to transmit information, and microwaves used to heat your lunch.

In this experiment, we take a closer look at the latter type of radiation: microwaves. Their wavelengths range from one meter to one millimeter, giving them a wide variety of applications. Furthermore, microwaves play a significant role in our understanding of the universe through the study of the cosmic microwave background, which is believed to be a remnant of the Big Bang, the event that marked the beginning of the universe. Microwaves exhibit many interesting properties. In the following, several of these properties will be examined and discussed, namely the angular distribution and polarization properties of microwaves, their wavelengths, and the formation of evanescent waves. Using these phenomena, we will also determine the lattice constant of a macroscopic cube.

2 Angular distribution of microwaves

To examine the angular distribution of microwaves, the transmitter was placed on a goniometertable, such that the horn is located approximately at the pivot point. The angular distribution was measured to both sides $\pm 90^\circ$ in equidistant steps. The measurements are done with a photo diode that is connected to an Agilent 34402A. The measured intensity was normalized to 1 for $\theta = 0^\circ$ and is shown in fig. 1. The density of measured points around $\theta = 0^\circ$ is quite low, which is not ideal. One should see the symmetry around the 0-point, but with our measurements, one can only guess it.

Nevertheless was the angle stretch of the radiation quantified by calculating the halfangle θ_H where the full width at half maximum lies.

$$\theta_H = (28.4 \pm 2.5)^\circ \quad (1)$$

As expected due to the transmitter being a horn antenna, this confirms that the microwave radiation is pointed towards the 0° direction.

3 Focal length of paraffin wax lens

To further investigate microwaves, it is necessary to produce collimated rays. For that a paraffin wax lens

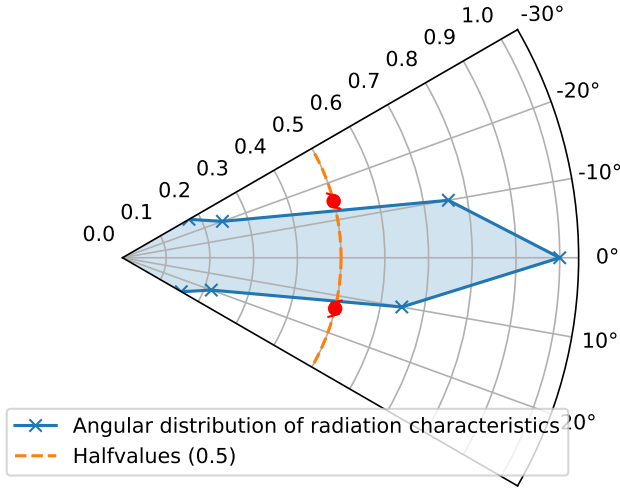


Figure 1: Measured voltage within the relevant angular range normalized to 1 in dependence of the angle θ . The left and right full width at half maximum are marked.

was used which focal point has to be determined. This was achieved by placing a wax lens between transmitter and receiver. The distance between the transmitter and the lens is $g = (80 \pm 1) \text{ cm}$. The receiver is placed with maximal possible distance to the lens and then progressively moved towards it. The radiation was measured in every step. To determine the peak radiation level, a smooth curve was fitted to the experimental data and its maximum was determined to $b' = (151.8 \pm 6.0) \text{ cm}$. The plot is shown in fig. 2.

The focal length was then calculated using the thin-lens equation:

$$f = \frac{g \cdot b}{g + b} \quad \text{with} \quad b = b' - g \quad (2)$$

which results in a focal length of ($f = 37.84 \pm 0.48 \text{ cm}$). In the following parts of the experiment, the focal length was roughly placed at 40 cm, such that the rays would be collimated.

4 Determination of the wavelength via standing waves

The wavelength of the microwave radiation was determined using standing waves. For this purpose, the positions of several nodes were measured along the optical bench. Since the distance between two neighboring nodes corresponds to half the wavelength, the wavelength can be calculated using

$$\lambda = 2\Delta x, \quad (3)$$

where Δx is the mean distance between adjacent nodes.

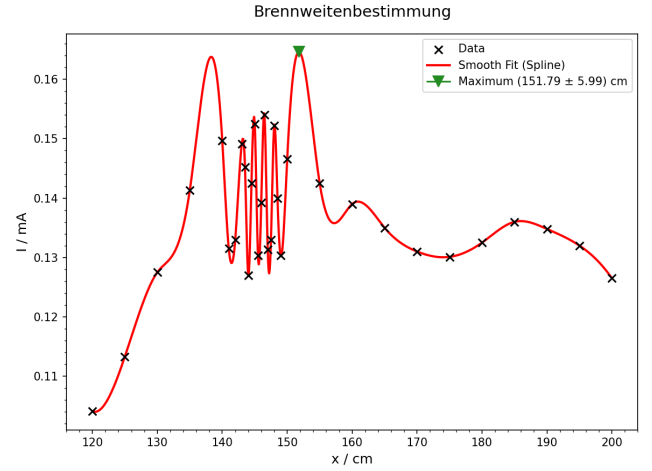


Figure 2: Intensity profile of the microwave radiation in relation to its distance from the lens.

From the measured values, an average node spacing of

$$\Delta x = (1.628 \pm 0.026) \text{ cm} \quad (4)$$

was obtained. This results in a wavelength of

$$\lambda = (3.256 \pm 0.052) \text{ cm}. \quad (5)$$

The experimentally determined wavelength is in good agreement with the expected wavelength of the microwave source.

5 Determination of the wavelength via the Michelson interferometer

The wavelength of the microwave radiation was additionally determined using a Michelson interferometer. By changing the position of one of the mirrors, the optical path difference between the two partial beams changes, resulting in alternating constructive and destructive interference.

The wavelength can be determined from the average distance between neighboring interference maxima according to

$$\lambda = 2\Delta x, \quad (6)$$

where Δx denotes the mean distance between adjacent maxima.

From the measured values, an average distance of

$$\Delta x = (1.650 \pm 0.050) \text{ cm} \quad (7)$$

was obtained. This results in a wavelength of

$$\lambda = (3.300 \pm 0.100) \text{ cm}. \quad (8)$$

The experimentally determined wavelength agrees within its margin of error with the wavelength obtained from the standing wave measurement.

6 Total reflection and frustrated total reflection

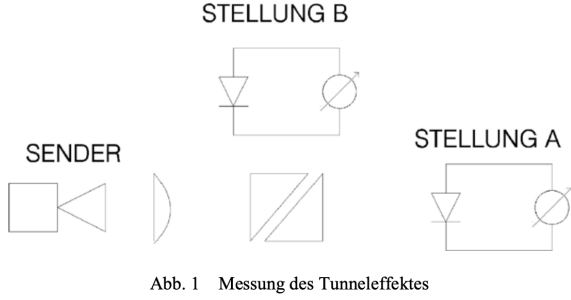


Figure 3: The setup to observe total reflection. With the two receivers A and B the reflected and transmitted intensity is measured.

In this part of the experiment, frustrated total reflection was investigated using two paraffin prisms separated by an air gap (fig.3). Although total reflection occurs at the interface, a part of the electromagnetic wave can penetrate through the gap due to the evanescent field. This behavior is analogous to the quantum mechanical tunneling effect.

The transmitted intensity was measured as a function of the prism separation d . An exponential decrease of the transmitted intensity is expected and can be described by

$$I_{transmitted}(d) = I_0 \exp\left(-\frac{d}{s}\right), \quad (9)$$

where I_0 is the initial intensity and s is the penetration depth of the evanescent field.

The measured data were fitted using the exponential function above. The fit yielded a penetration depth of

$$s = (2.31 \pm 0.14) \text{ cm}. \quad (10)$$

The measured transmission intensity decreases exponentially with increasing prism separation, while the reflected intensity increases correspondingly. This behavior confirms the existence of the evanescent wave and demonstrates the tunneling effect for microwaves.

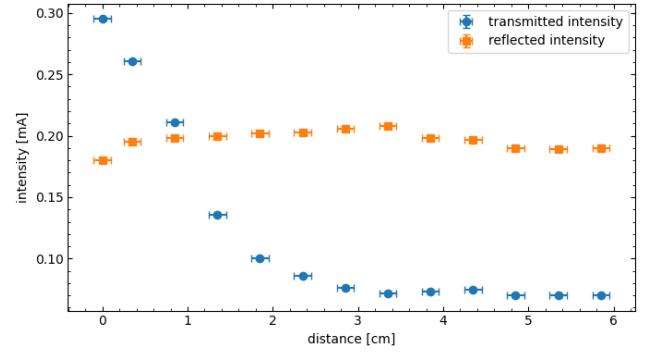


Figure 4: The measured transmitted and reflected intensity. As one grows the other shrinks.

7 Polarisation of microwaves

In this part of the experiment, the polarization properties of microwave radiation were investigated using a metallic grid placed between transmitter and receiver. The grid acts as a polarization filter since electric field components parallel to the metallic wires are suppressed, while perpendicular components are transmitted.

The transmitted intensity was measured as a function of the rotation angle θ of the grid. The intensity shows a clear angular dependence, which demonstrates that the emitted microwave radiation is linearly polarized.

For an ideal polarization filter, the transmitted intensity follows Malus' law. Since both the grid and the receiver contribute to the polarization effect, the measured intensity can approximately be described by

$$I(\theta) \propto \cos^4(\theta). \quad (11)$$

The measured intensities were normalized to the maximum intensity and displayed in a polar plot (fig.5). The intensity reaches its maximum for parallel alignment and decreases significantly for perpendicular orientations.

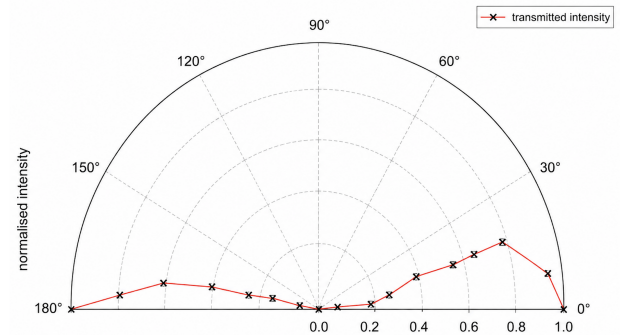


Figure 5: Polar plot of the normalized transmitted intensity as a function of the grid angle.

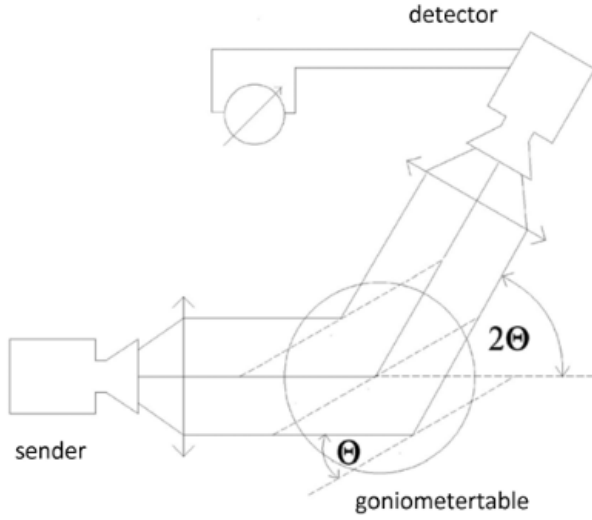


Figure 6: Experimental setup to measure the lattice constant of an macroscopic cube

The observed angular dependence confirms the linear polarization of the microwave radiation emitted by the Gunn diode.

8 Determination of the lattice constant of a cubic crystal

In solid state physics, the θ -rocking-curve method is often used to examine the characteristics of an atomic lattice, namely the lattice constant. In this part of the experiment, this method was tested with microwaves on a macroscopic styrofoam cube doted with aluminum wrapped balls, such that a cubic lattice emerges. The bragg's law states that monochromatic radiation of wavelength λ interferes constructive under the angle θ if

$$n\lambda = 2d\sin(\theta) \quad (12)$$

is realised. Here n is the number of the maximum or diffraction order, λ the wavelength measured in a previous part, d the spacing between adjacent crystal planes, and θ is the angle at which the maximum occurs. The setup used in the experiment is shown in fig. 6. Starting the measurement at an angle $\theta = 0^\circ$, the crystal was rotated in steps of 2° while the detector was rotated to be at 2θ to ensure the symmetric condition required for Bragg reflection.

While Bragg's law is independent of the orientation of the crystal planes, it acutally plays a role in the measurement. That is why we need to introduce Miller indices h, k, l to describe which plane we are currently measuring. For a cubic crystal, the spacing between

lattice plane can be expressed as:

$$d_{hkl} = \frac{d_0}{\sqrt{h^2 + k^2 + l^2}} \quad (13)$$

Using equations 12 and 13 one can calculate the lattice constant d_0 with:

$$d_0 = d_{hkl} \sqrt{h^2 + k^2 + l^2} = \frac{n\lambda}{2\sin(\theta)} \sqrt{h^2 + k^2 + l^2} \quad (14)$$

The sides with the Miller indices (100) and (110) were measured and the intensity in terms of the rotation angle is shown in fig. 7 and 8 respectively.

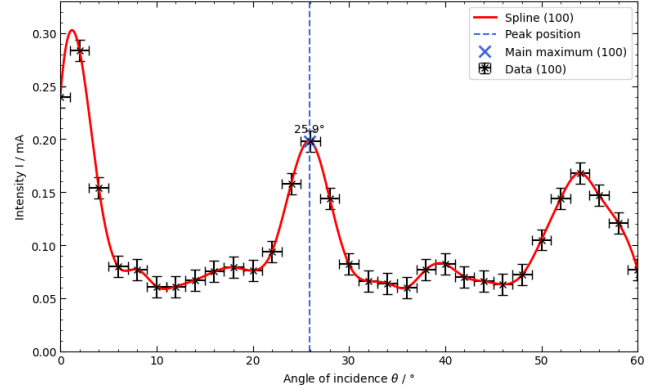


Figure 7: Measured Bragg diffraction from the (100) plane for angles $\theta = [0, 60]$. The first maximum was determined and marked.

The first maximum for the (100) plane was measured at an angle $\theta = (25.89 \pm 0.5)^\circ$.

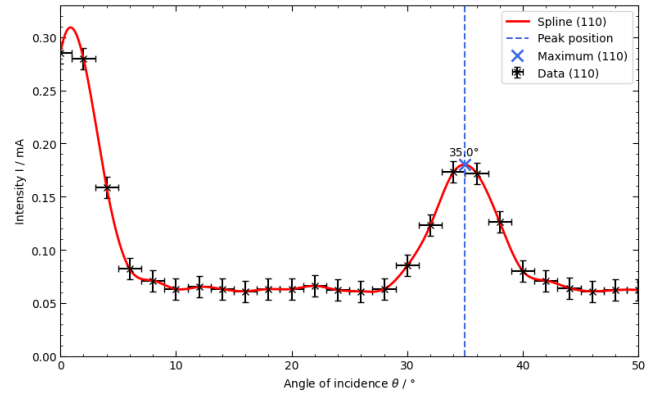


Figure 8: Measured Bragg diffraction from the (110) plane for angles $\theta = [0, 50]$. The first maximum was determined and marked.

The first maximum for the (110) plane was measured at an angle $\theta = (34.95 \pm 0.5)^\circ$.

Using eq. 14 and the wavelength $\lambda = \text{hehrher}$ the lattice constant can be calculated:

$$\begin{aligned} (100) : d_0 &= (3.74 \pm 0.14) \text{ cm} \\ (110) : d_0 &= (4.03 \pm 0.10) \text{ cm} \end{aligned} \quad (15)$$

Both values for the lattice constant are in a reasonable order of magnitude considering the dimension of the styrofoam cube, but aren't consistent within their uncertainties. The deviation could have its origin in the 80's style experimental setup where the cube was loosely placed on the goniometric table and therefore not perfectly aligned.

None the less this experiment shows that this method of examining an unknown lattice structure is working and with numerous alternations of the setup e.g. a more modern type, it can yield reliable data.

9 Properties of a circular waveguid

Waveguides are resonators with two open ends, allowing waves to propagate on its interior from one end to the other with minimal loss of energy by restricting the transmission of energy to only the direction of the guide. There are three types of waveguides: acoustic waveguides which direct sound, optical waveguides which direct light, and radiofrequency waveguides which direct electromagnetic waves other than visible or near visible light.

In this part of the experiment one of the last type was used to guide microwaves. It is a hollow conductive metal pipe which inner walls are often plated with a material which reflects signals as much as possible such that the loss is reduced. For high end waveguides silver or gold is often used. It works by trapping and "bouncing" the waves off its highly conductive inner walls, directing the energy forward. With that there can be two different propagation modes, one where the electric field is completely perpendicular to the direction the wave is traveling and one where the magnetic field is completely perpendicular to the travel direction of the wave.

To investigate this a transmitter was placed at one, a receiver was placed at the other end of a waveguide. Despite the curvature and length of the guide a clear signal was detected. To verify that the signal was transmitted through the interior of the pipe and not just on its surface, we closed one end with a metal cap. Since no signal, besides the background noise could be detected, it is proof to the previous claim.

Finally we examined the polarization of the transmitted waves by rotating the receiver. A strong dependence of the measured signal to the rotation of the receiver was observed, signaling that the wave leaving the waveguide is indeed polarized. The maximum was observed at a rotation angle of approximately 90° relative to the transmitter. This indicates that the polarization direction was altered during propagation through the waveguide.

10 Conclusion

In this experiment, several fundamental properties of microwave radiation were investigated successfully. The measurements demonstrated characteristic wave phenomena such as interference, diffraction, polarization, and total reflection. Furthermore, the experiments showed that many principles known from optics can also be observed for microwaves.

Different experimental methods were used to determine the wavelength of the emitted microwave radiation. The obtained results were consistent and confirmed the reliability of the measurements and the experimental setup.

The investigation of frustrated total reflection demonstrated the existence of evanescent waves and illustrated the microwave analogue of the quantum mechanical tunneling effect. In addition, the polarization measurements confirmed that the emitted microwave radiation is linearly polarized.

Using Bragg diffraction on a macroscopic crystal structure, the lattice constant of the artificial crystal could be determined. Although small deviations occurred due to experimental uncertainties and alignment errors, the measured values were in reasonable agreement with the expected dimensions of the structure.

Finally, the experiments with the circular waveguide demonstrated the efficient transmission of microwaves through conductive structures and showed the influence of the waveguide geometry on the polarization properties of the transmitted radiation.

Overall, the experiment provided a comprehensive overview of important properties of microwave radiation and demonstrated several physical effects that are highly relevant in optics, solid state physics, and modern communication technologies.

References

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